

What is Remote Browser Isolation (RBI)?

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Remote Browser Isolation (RBI) is a security technology that isolates web browsing activity from a user's device, typically by running the browser in a remote environment and streaming the visual output to the user. This approach protects against web-based threats by ensuring that any malicious content is contained within the remote environment and does not reach the user's device.

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What is remote browser isolation?

The internet is simultaneously the most important productivity tool for a business and its biggest liability due to the wide range of security threats it exposes them to. While it is possible to protect against some web-browsing threats using old techniques like blocking known risky domains, these precautions don't guard against other exploitations. So, how do organizations protect users, data, and systems from the ever-increasing threats on the web?

Remote browser isolation (RBI), also known as web isolation, is a security measure that separates users' devices from the act of internet browsing by hosting and running all browsing activity in a remote cloud-based container. This act of sandboxing internet browsing protects data, devices, and networks from all kinds of threats originating from infected website code, including:

- Web-borne malware and ransomware
- Zero-day exploits
- Browser vulnerabilities, such as plug-ins
- Infected file downloads
- Malicious web links in phishing emails
- And many others



RBI

Remote Browser Isolation



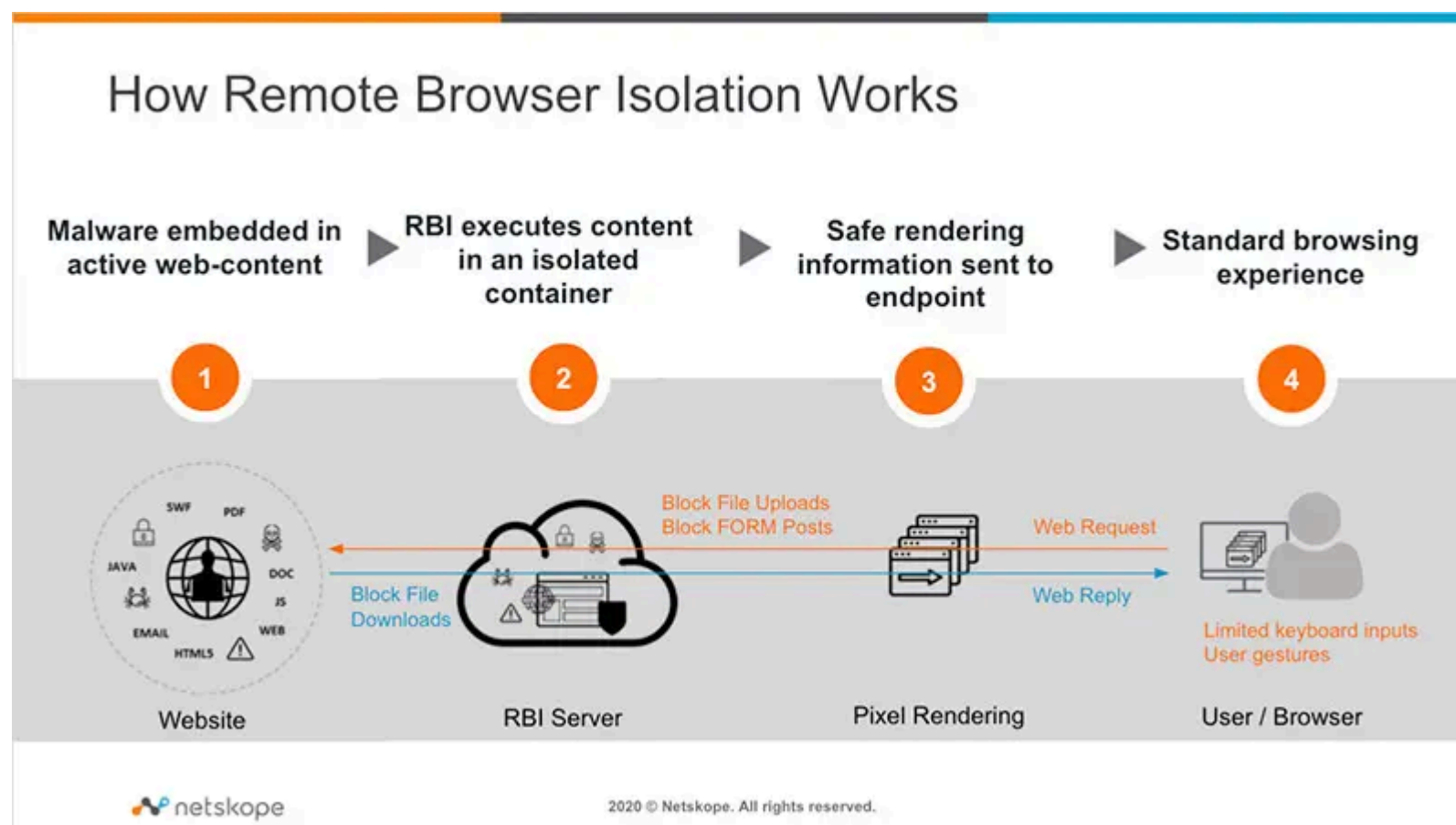
Also known as web isolation, a security measure that separates users' devices from the act of internet browsing by hosting and running all browsing activity in a remote cloud-based container.



How does remote browser isolation technology work?

Remote browser isolation adopts **Zero Trust principles** and **applies them to the act of internet browsing**. Instead of deciphering between good and bad web browsing, **remote browser isolation determines that targeted websites are not trusted and thus isolated in a container where no website code executes on endpoints**. Let's dive a little deeper into the workings of RBI and how it protects users from threats:

1. **Website exists with corrupted code** – A threat is programmed into the code of a website which threatens anyone browsing to that particular site.
2. **RBI isolates web content** – Since the particular **user has remote browser isolation**, all web pages are hosted and rendered on a remote container separated from the user's endpoint device (computer, smartphone, tablet, etc.)
3. **RBI renders content to the user** – The **user is able to view a pixel rendering of the web page via the remote web browser executing in an isolated container**.



As far as the user is concerned, the experience is the same as using a standard browser. The only difference is peace of mind.

Users enjoy all of the standard benefits of internet browsing without the threats of exposing devices to the raw web code, just like viewing a tiger through a fence versus watching the tiger while inside the cage. You still see the tiger while you're protected by a barrier.

Datasheet: [Netskope Remote Browser Isolation](#)

Learn More: [What is Zero Trust Security?](#)

What are the remote browser isolation rendering modes?

There's more than one way to render web pages using remote browser isolation and some are more secure than others. These are the different RBI rendering modes:

1. Pixel rendering

This is the ideal RBI rendering mode for high-risk users such as executives and administrators with a wide array of access to sensitive data, assets, and infrastructures. Pixel rendering allows no website content (pictures, fonts, style sheets, JavaScript, etc.) to interact with the endpoint browser or operating system. All web browsing is completely sealed off from the user and is a good choice for high-risk websites.

2. DOM-based rendering

This is the next step down from pixel rendering. DOM (document object module)-based rendering isolates risky web-page elements, such as JavaScript, using RBI to pixel stream those risky elements to the user. However, other less-risky elements like fonts and style sheets are rendered by the user's local browser. This method is used in order to streamline the RBI process and leverages the processing power of the user's endpoint device while still providing security where it is most needed. This is a good choice for medium-risk websites.

3. Streaming media

This RBI option will pixel render risky web page elements, allow safe DOM elements to render in the user's web browser, and streaming media is natively provided to the browser without pixel rendering for a better user experience. This is a good choice for lower-risk streaming media websites where the user experience is important.

What are the different types of remote browser isolation and their use cases?

1. **Targeted RBI** – Couples with secure web gateways to protect against uncategorized and risky websites. Creates disposable remote browser containers when the system detects these risky URLs. This is the most popular form of RBI.
2. **Full RBI** – Best for high-risk users such as C-suite and executive-level users with access to sensitive information. All websites are rendered using remote browser isolation.
3. **Email RBI** – Renders all embedded URLs in emails as "read-only" for the purposes of preventing phishing attacks.
4. **Document RBI** – All documents downloaded via the web browser are rendered in the cloud as a static PDF.
5. **Application Access Control RBI** – Allows untrusted users and devices to view or use pixel rendered applications or data without giving them the ability to alter these assets.

How does remote browser isolation fit within a SASE architecture?

Secure Access Service Edge (SASE) is a cloud-based architecture that converges networking and security tools meant to secure data, users, and applications. This is the future of cloud security that replaces the traditional on-premises perimeter-based architectures that we are all used to with appliances (access control lists, network firewalls, etc.). All of our networking and security is destined to move from on-location devices to the cloud.

But what about SSE? Security Service Edge (SSE) is the security services half of the SASE architecture, which is built to simplify your infrastructure into one single solution.

Blog: [Netskope Integrates Targeted RBI Within SASE Architecture](#)

Blog: [The Case for Targeted Remote Browser Isolation with Next Gen SWG](#)

Where does remote browser isolation fit in all of this?

While remote browser isolation is not a standalone solution within the SASE/SSE framework, it is a feature of the architecture and works alongside secure web gateways (SWG). Specifically, SWG is enhanced by targeted RBI to protect against threats from uncategorized and other risky websites. While the rest of SSE is focused on protecting your data, network, and cloud-based assets, remote browser isolation guards your user's web browsing experience. RBI is yet another piece of a zero trust-based security model that makes up the SASE security philosophy.

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How to Protect your Web Browsing Strategy with RBI (In Spanish)



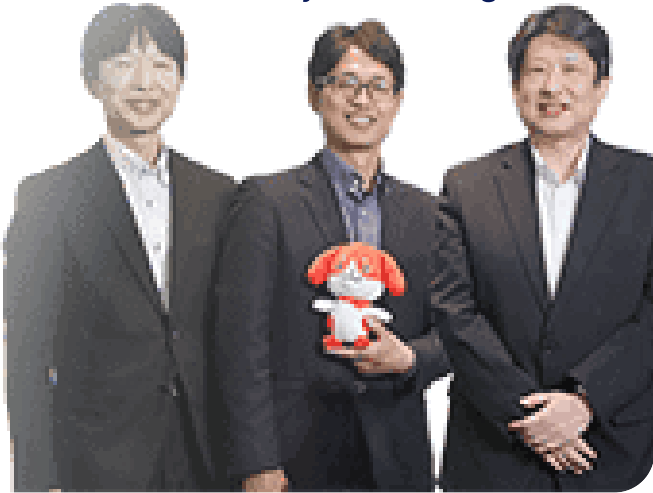
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